

Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution Affiliated to Anna University)

Machine Learning Algorithms Laboratory

Experiment No.	8
Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN and Hierarchical Clustering
Subject Code	ICS1512
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1 Aim and Objective

To cluster human activity recognition (HAR) sensor data using K-Means, DBSCAN and Hierarchical (Agglomerative) clustering; select hyperparameters via elbow/silhouette and density search; and compare algorithms using internal and external validation metrics.

2 Dataset

- **Name:** UCI Human Activity Recognition Using Smartphones
- **Features:** 561 time- and frequency-domain signals from accelerometer and gyroscope
- **Activities:** WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING
- **Preprocessing:** StandardScaler + PCA (20 components $\approx$ 77.5% variance for clustering; 2 components for visualisation)
- **Analysis sample:** Stratified 3000 training samples (500 per activity) for computational feasibility

3 Theory

K-Means partitions data into k spherical clusters by minimising within-cluster sum of squares. k is chosen via elbow curve and silhouette score.

DBSCAN finds density-connected regions; parameters `eps` and `min_samples` control neighbourhood size. Points that are not dense are labelled as noise (-1).

Hierarchical (Agglomerative) builds a tree of merges (linkage: ward, complete, average, single). Number of clusters is obtained by cutting the dendrogram.

4 Exploratory Analysis

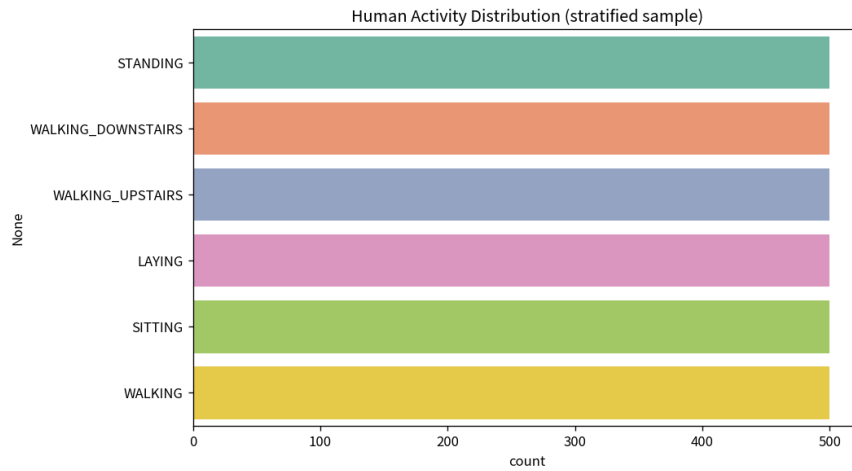


Figure 1: Activity class distribution (stratified sample).



Figure 2: PCA projection coloured by true activity labels.

Inference: Static activities (LAYING, SITTING, STANDING) separate from dynamic locomotion activities in PCA space, suggesting a natural two-group structure.

5 K-Means

k	WCSS	Silhouette
2	588084	0.4936
3	476075	0.4138
4	434312	0.3585
5	396377	0.2144
6	374519	0.2071
7	356519	0.1715
8	339947	0.1814

Table 1: K-Means elbow and silhouette results.

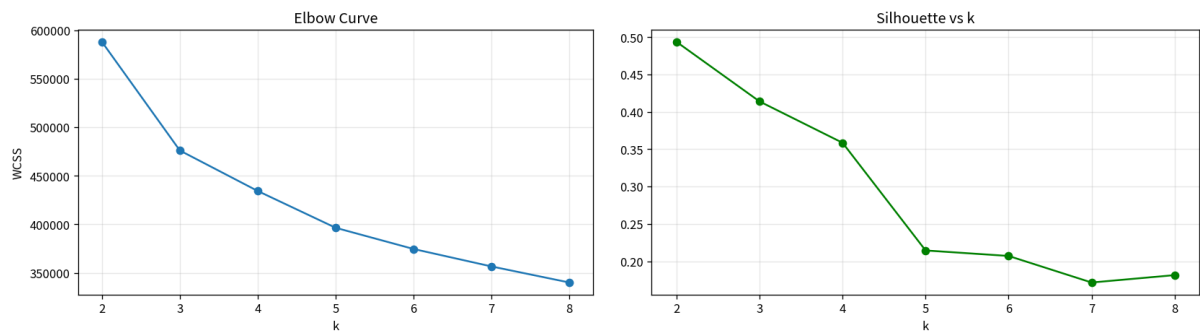


Figure 3: Elbow curve and silhouette scores.

Selected $k = 2$ (highest silhouette). This aligns with static vs dynamic activity separation.

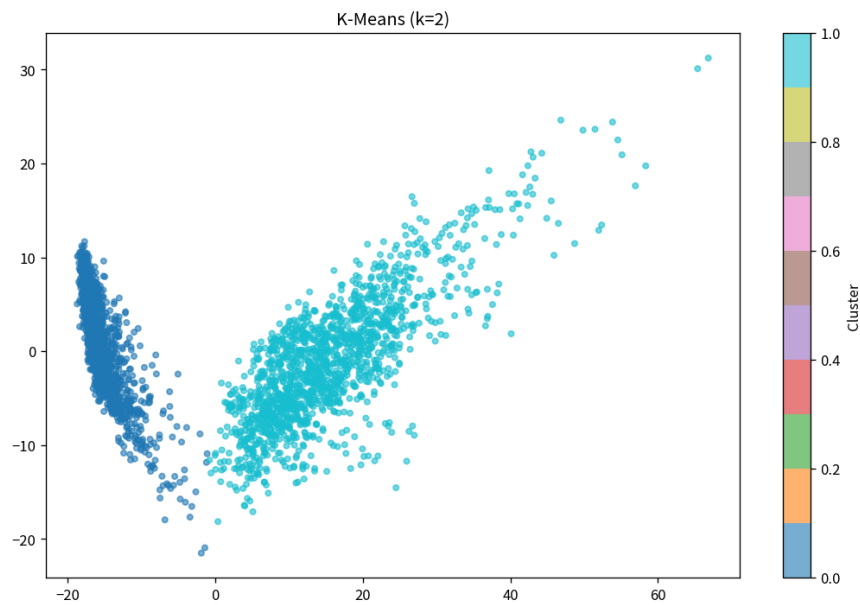


Figure 4: K-Means clusters in PCA space.

6 DBSCAN

Parameter search over $\text{eps} \in \{0.5, \dots, 2.0\}$ and $\text{min_samples} \in \{5, 10, 15\}$ on 2-D PCA.

- Best: $\text{eps} = 0.7$, $\text{min_samples} = 15$
- Clusters found: 7 (plus substantial noise)
- Silhouette (non-noise): 0.754

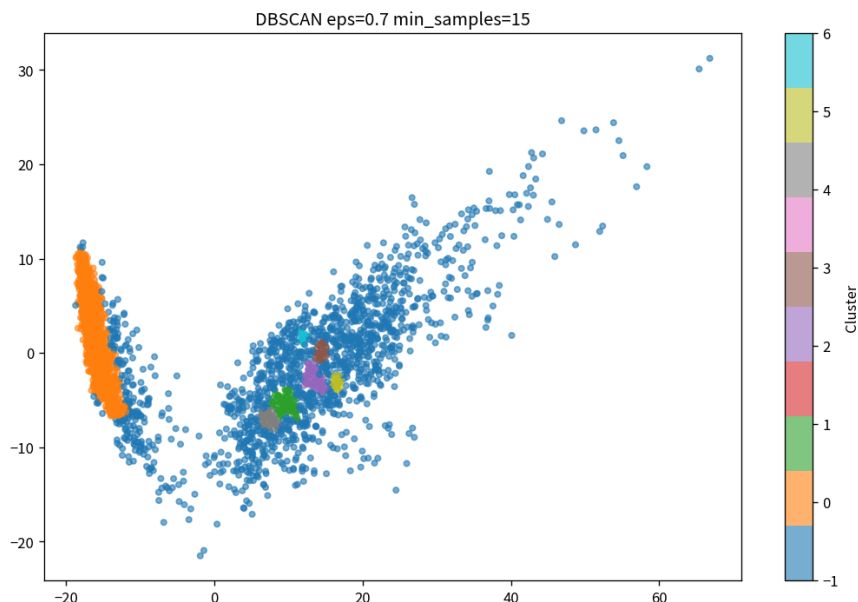


Figure 5: DBSCAN clusters (noise points included).

7 Hierarchical Clustering

Ward linkage with $k = 2$; dendrogram computed on the sample.

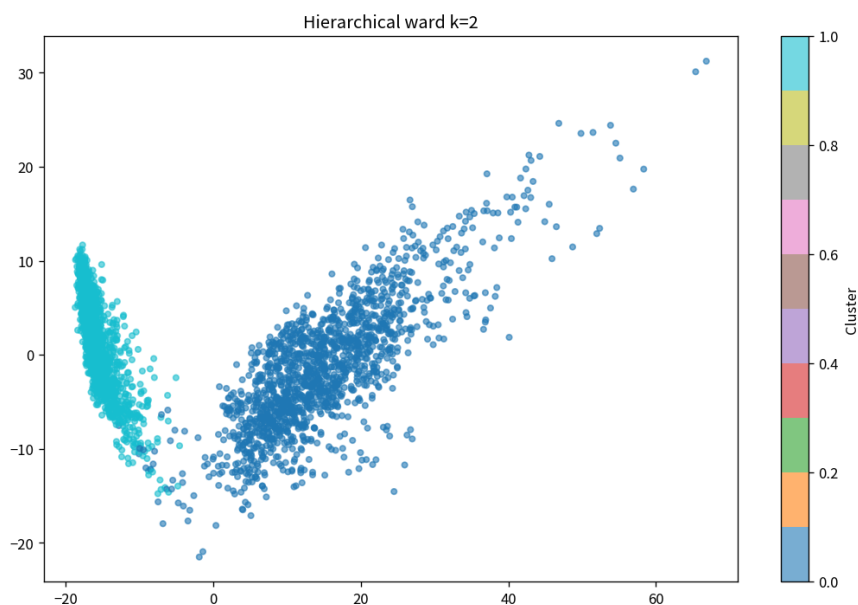


Figure 6: Hierarchical (ward) clusters.

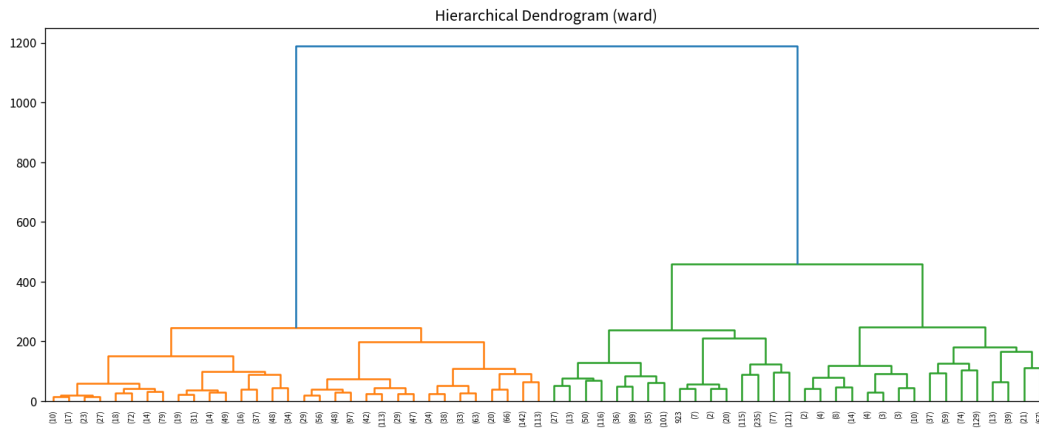


Figure 7: Dendrogram (ward, truncated).

Linkage	Silhouette	DB	ARI	NMI
ward	0.490	0.810	0.319	0.518
complete	0.468	0.880	0.002	0.040
average	0.559	0.347	0.000	0.001
single	0.559	0.347	0.000	0.001

Table 2: Linkage method comparison ($k = 2$).

Ward linkage best matches true activity structure (highest ARI/NMI).

8 Final Metrics Comparison

Algorithm	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
K-Means	0.494	0.808	3652	0.329	0.545
DBSCAN	0.754	0.527	1991	0.222	0.354
Hierarchical	0.490	0.810	3554	0.319	0.518

Table 3: Internal and external validation metrics.

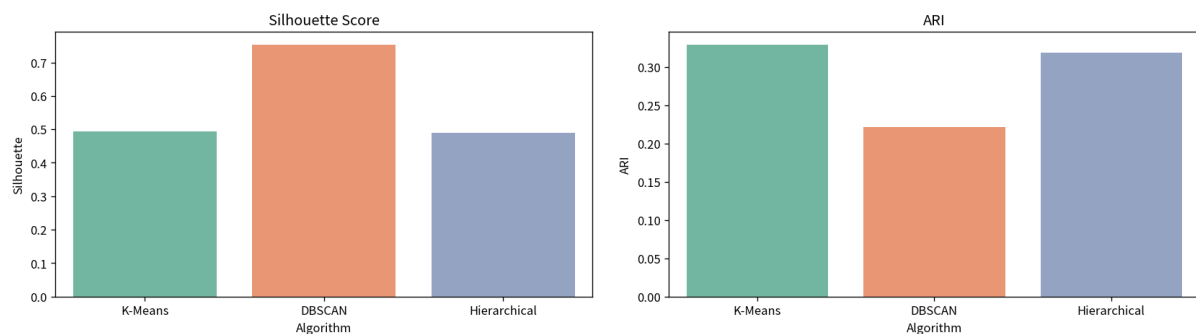


Figure 8: Silhouette and ARI comparison.

9 Cluster–Activity Crosstab (K-Means)



Figure 9: K-Means cluster membership vs true activity.

Cluster 0 captures static activities (LAYING/SITTING/STANDING); Cluster 1 captures walking variants — consistent with the silhouette-selected $k = 2$.

10 Analysis

- HAR data is naturally bipartite (static vs dynamic). Forcing $k = 6$ (true class count) yields lower silhouette.
- K-Means and Hierarchical (ward) achieve the best external agreement (ARI/NMI).
- DBSCAN obtains the highest internal silhouette on dense regions but labels many points as noise and has weaker ARI against the six activity labels.
- Ward linkage is clearly superior to single/average/complete for this dataset.

11 Conclusion

K-Means with $k = 2$ is the most practical model for this HAR feature set when the goal is coherent, interpretable groups. Hierarchical ward clustering performs similarly. DBSCAN is useful for detecting dense cores and outliers but does not recover the six semantic activities cleanly. PCA reduction is essential for density-based methods and visualisation.

12 Key Learning Outcomes

- Elbow and silhouette methods for choosing k
- DBSCAN parameter search (`eps`, `min_samples`) and noise handling

- Linkage comparison in hierarchical clustering
- Internal (Silhouette, DB, CH) vs external (ARI, NMI) evaluation

13 References

1. D. Anguita et al., UCI HAR Dataset, 2013.
2. MacQueen, “Some methods for classification and analysis of multivariate observations,” 1967.
3. Ester et al., “A density-based algorithm for discovering clusters in large spatial databases,” KDD 1996.
4. F. Pedregosa et al., Scikit-learn, JMLR 2011.

— *End of Report* —